Development Plan: Inventory Management System (IMS) for Thrifted & Branded Surplus Clothing (with Item Approval Workflow)

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This plan covers Week 4 (September 7) to Week 14 (November 22), 2026, from project planning and requirements analysis through design, development, system testing, acceptance testing, defect fixing, deployment, and final presentation. Development begins in Week 8 (October 5), after the database and interface design work is complete, and is organized by workflow so the intake and approval modules are built before the sales, stock, and reporting modules that depend on them. System testing is automated with a Postman collection run through Newman and is followed by role-based acceptance testing, in which testers play the Staff, Reviewer, and Admin roles. Defects from either round are fixed and re-tested before deployment. Dates follow the accompanying Gantt chart.

2.1.1 Timeline and Schedule

Week	Date	Development Activity	Major Tasks / Deliverables
Week 4	Sep 7–13	Project Planning and Documentation (WBS 1.1)	Finalize project scope; draft Development Plan and Test Plan; process flow and analysis
Week 5	Sep 14–20	Requirements Analysis (WBS 1.2)	Gather Staff, Reviewer, and Admin needs; document functional and non-functional requirements
Week 6	Sep 21–27	System Analysis and Design (WBS 2.1)	Finalize item status flow; define user roles and permissions; use cases and system design
Week 7	Sep 28–Oct 4	Database and Interface Design (WBS 2.2)	Database design; UI layouts for Login, Item Form, and Pending Queue
Week 8	Oct 5–11	System Development – Item Intake & Authentication (WBS 3.1)	Project setup, database tables, and Login screen; role-based access control; Add Item form (new items set to Pending)
Week 9	Oct 12–18	System Development – Approval Workflow (WBS 3.2)	Pending Items queue; Approve/Reject actions with reason capture; status transition logic

Week	Date	Development Activity	Major Tasks / Deliverables
Week 10	Oct 19–25	System Development – Sales & Stock (WBS 4.1); Reports & Integration (WBS 4.2)	Sale/checkout screen; stock tracking and low-stock flagging; In Stock-only sales rule; inventory search and dashboard; reports screens; module integration and status logging
Week 11	Oct 26–Nov 1	System Testing – Postman Collection Automation (WBS 5.1)	Postman collection and test cases; Newman test runs; defect logging and test report
Week 12	Nov 2–8	Acceptance Testing – Role-Based Walkthrough (WBS 5.2)	Acceptance testing with testers in the Staff, Reviewer, and Admin roles; tester feedback and usability evaluation; acceptance test report and sign-off
Week 13	Nov 9–15	Defect Fixing, Final Testing, and Deployment (WBS 6.1)	Defect fixing and refinement; start of regression and final testing
Week 14	Nov 16–22	Final Testing, Deployment, Documentation, and Presentation (WBS 6.1–6.2)	Regression and final testing; deployment; final documentation and presentation of the deployed system

2.1.2 Process Flow & System Design

Item status transition process flow

1. Staff logs in and opens the Add Item form.
2. Staff enters the brand, size, condition, price, and photo and submits the form. The system validates the input and saves the item with the status "Pending".
3. The item appears in the Pending Items queue, and Staff cannot sell it at this stage.
4. The Reviewer logs in, opens the Pending Items queue, and physically inspects the garment for quality, physical condition, tags, and authenticity.
5. The Reviewer records the result in the system:
    - Approve: the status changes from "Pending" to "In Stock".
    - Reject: the Reviewer must enter a rejection reason; the status returns to "Pending" and the reason is recorded against the item.

6. Every decision is written to the status log with the item, previous status, new status, user, and timestamp.
7. Staff searches for an "In Stock" item and processes the sale on the Sale/Checkout screen. The Status Guard Rule blocks the sale of any item that is not "In Stock".
8. On a successful sale, the status changes from "In Stock" to "Sold", the stock count is reduced, and the change is logged.
9. The Admin searches the inventory, monitors stock levels and low-stock alerts, and generates status reports from the logged data.

UI design screens to be created

- Login screen
- Add Item form
- Pending Items queue (with Approve and Reject actions and the rejection reason field)
- Inventory list / dashboard (with search and low-stock indicators)
- Sale / checkout screen
- Reports screen

2.1.3 Gantt Chart Breakdown

The Gantt chart and work breakdown structure (WBS) are maintained in the accompanying Excel workbook, IMS Thrifted & Branded Surplus Gantt Chart (.xlsx). It holds the full WBS (levels 1 to 6.2.3), task IDs, assigned roles, start and end dates, durations, status, and the week-by-week bars for Weeks 4 to 14 (September 7 to November 22, 2026). This plan follows the workbook's dates, and the workbook remains the place where the schedule is updated.

The workbook organizes the work into six phases. Each phase starts and ends on a week boundary (Monday to Sunday), and Phase 4 runs for one week.

WBS	Phase	Weeks	Dates
1	Planning & Requirements	Weeks 4–5	Sep 7–20
2	Analysis & Design	Weeks 6–7	Sep 21–Oct 4
3	Intake, Authentication & Approval	Weeks 8–9	Oct 5–18
4	Sales, Stock & Reports	Week 10	Oct 19–25
5	Testing (System Testing & Acceptance Testing)	Weeks 11–12	Oct 26–Nov 8
6	Finalization & Deployment	Weeks 13–14	Nov 9–22

Each phase is divided into these work packages:

WBS	Work Package	Task ID	Assigned To	Dates
1.1	Project Planning & Documentation	IMS-01-01	Project Manager & System Analyst	Sep 7–13

WBS	Work Package	Task ID	Assigned To	Dates
1.2	Requirements Analysis	IMS-01-02	System Analyst	Sep 14–20
2.1	System Analysis & Design	IMS-02-01	System Analyst & UI/UX Designer	Sep 21–27
2.2	Database & Interface Design	IMS-02-02	System Developer & UI/UX Designer	Sep 28–Oct 4
3.1	Item Intake & Authentication	IMS-03-01	System Developer	Oct 5–11
3.2	Approval Workflow	IMS-03-02	System Developer	Oct 12–18
4.1	Sales & Stock	IMS-04-01	System Developer	Oct 19–22
4.2	Reports & Integration	IMS-04-02	System Developer	Oct 22–25
5.1	System Testing (Postman Collection Automation)	IMS-05-01	QA & Tester	Oct 26–Nov 1
5.2	Acceptance Testing (Role-Based Walkthrough)	IMS-05-02	QA, Tester & System Analyst	Nov 2–8
6.1	Defect Fixing, Final Testing & Deployment	IMS-06-01	QA & System Developer	Nov 9–19
6.2	Documentation & Presentation	IMS-06-02	Project Manager & System Analyst	Nov 19–22

Each task in the workbook is color-coded by status (green for Complete, blue for In Progress, grey for Not Started), and the status and assignee columns use dropdown lists so the team can keep the chart current.